Car Sales Analysis
Programming for Data Science & AI
Philadelphia University
Second Semester 2024–2025
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Project Overview

•Objective: Analyze car sales data to identify trends and patterns

•Tools used: Python (Pandas and NumPy)

•Dataset: 2.5 million rows with cars, prices, commissions, and salespeople

• ☞*How I analyzed it:* gnisu tesatad eht dedaoL pandas.read_csv() htiw erutcurts derolpxe dna , ()ofni. dna ()daeh.

Dataset Description

- Rows: –2,500,000 Columns: 9
- Key columns:
  - Car Make, Car Model
  - Sale Price, Commission Rate, Commission Earned
  - Date, Car Year, Salesperson
- ☞*How I analyzed it:* desU df.shape , df.columns dna ,df.describe()

Data Cleaning

•Removed duplicates using df.drop_duplicates()

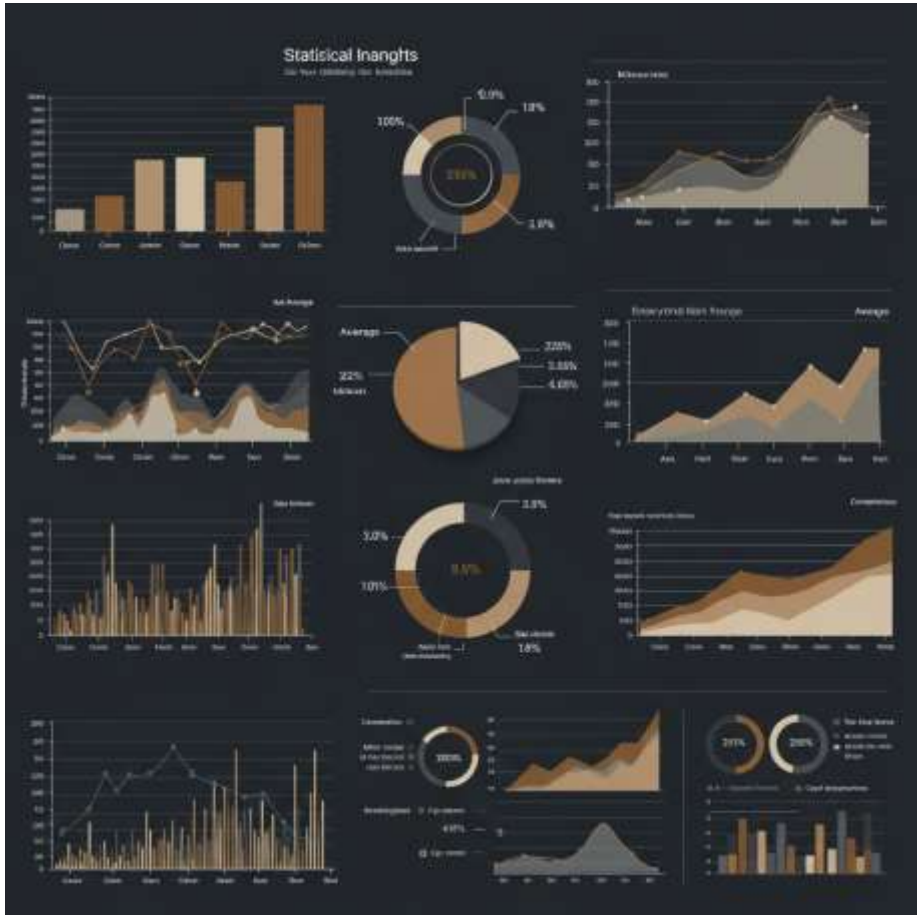
•Removed missing values using df.dropna()

•Converted Date gnisu emitetad ot pd.to_datetime()

•Created:

•Sale Year = Dateraey.td.

•Car Age =Sale Year - Car Year



Descriptive Statistics

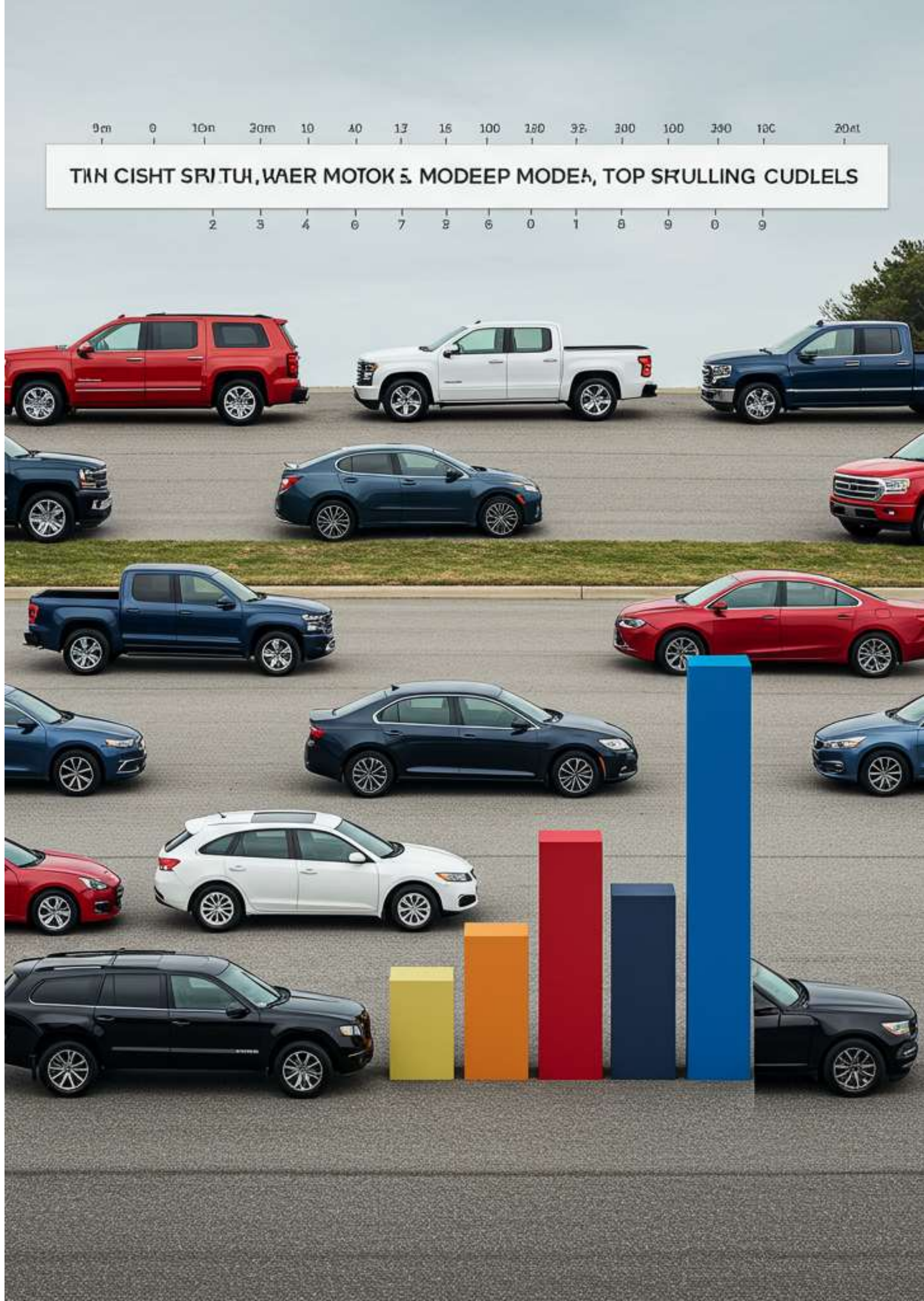
- **Average Sale Price:** 30,012 $
- **Average Commission:** 3,001 $
- **Price Range:** 50,000 –10,000
- **Commission Rate:** 15% –5%
- ☞*How I analyzed it:* desU df.describe() for numeric insights

Top-Selling Car Models

1. Silverado 15.04 –B
2. Corolla 15.01 –B
3. Civic 15.00 –B
4. F-14.98 –150 B
5. Altima 14.97 –B

- 👉*How I analyzed it:*desU

```
df.groupby('CarModel')['SalePrice'].sum().sort_values(ascen
ng=False)
```



Top Car Makes

1. **Honda** 15.03 – B
2. **Chevrolet** 15.02 – B
3. **Toyota** 15.00 – B
4. **Ford** 14.99 – B
5. **Nissan** 14.97 – B

- ☞*How I analyzed it:* no ybpuorg desU Car Make

Salesperson Performance

1. Kiara Mcdowell MD 7,488.91 –B
2. Dr. Christina Dennis 7,480.56 –B
3. Mr. Jon Bennett 7,474.75 –B

- 👉 *How I analyzed it:*

```
df.groupby('Salesperson')['Commission Earned'].mean()
```

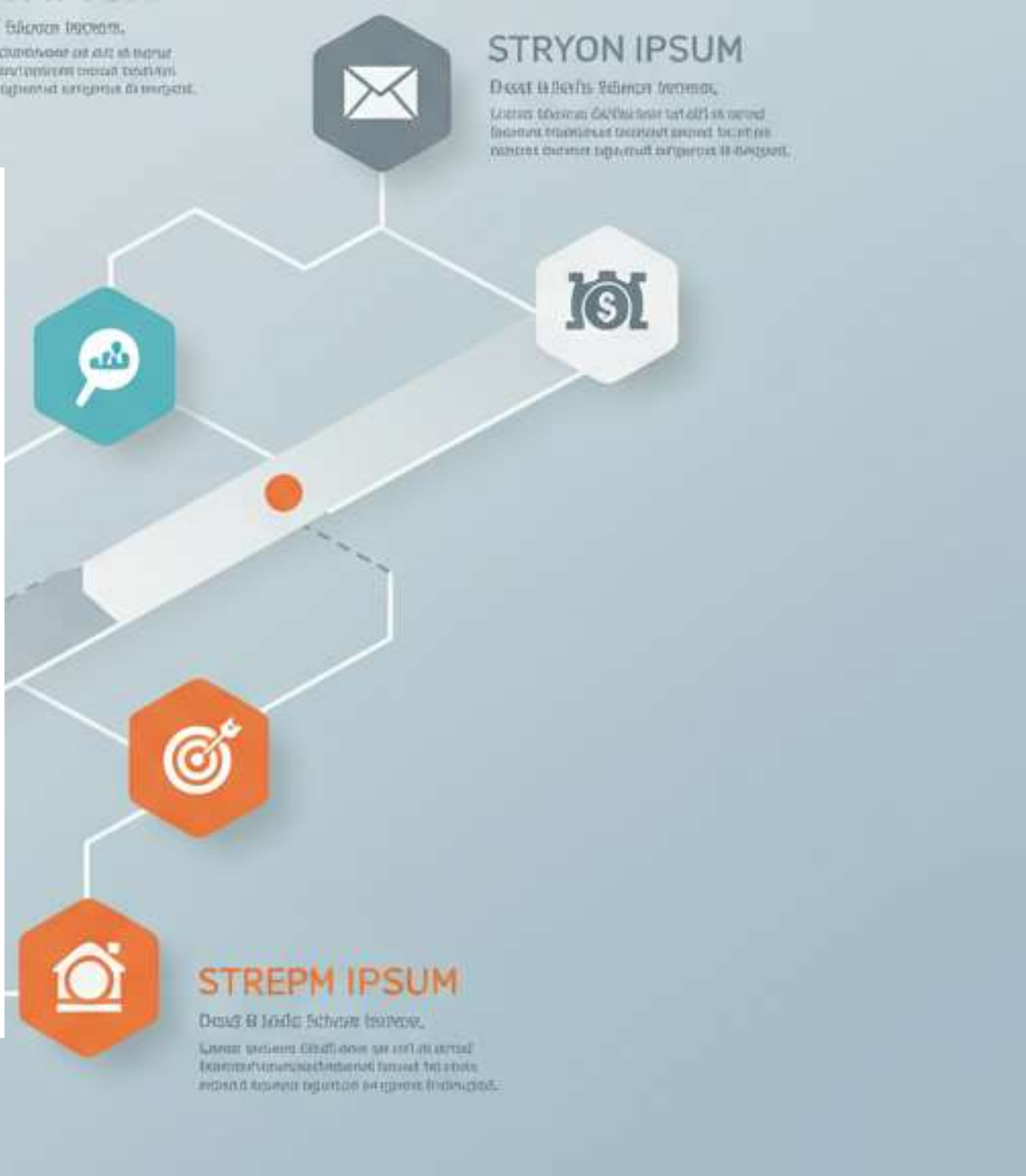
Correlation Between Price & Commission

- **Correlation coefficient: 0.78**
- Strong positive relationship
- ☞*How I analyzed it:*

```
np.corrcoef(df['Sale Price'], df['Commission Earned'])
```

Conclusions & Recommendations

- Focus on newer, more expensive cars
- Promote top-selling models like Silverado, Corolla
- Support top salespeople with incentives
- Use data-driven insights for strategy



Thank You?

ANY QUESTIONS?